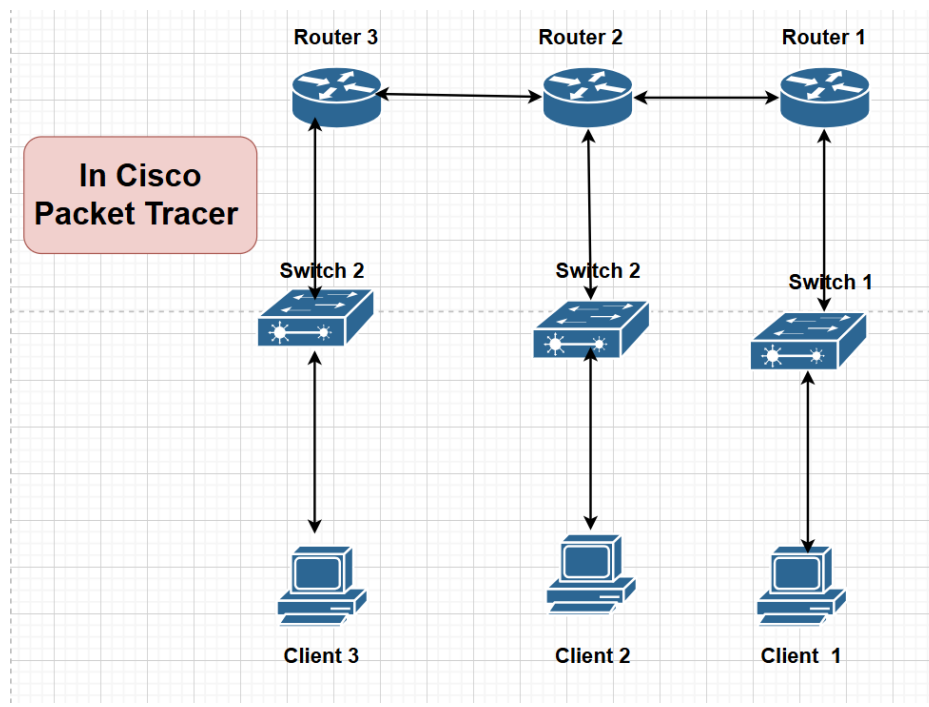
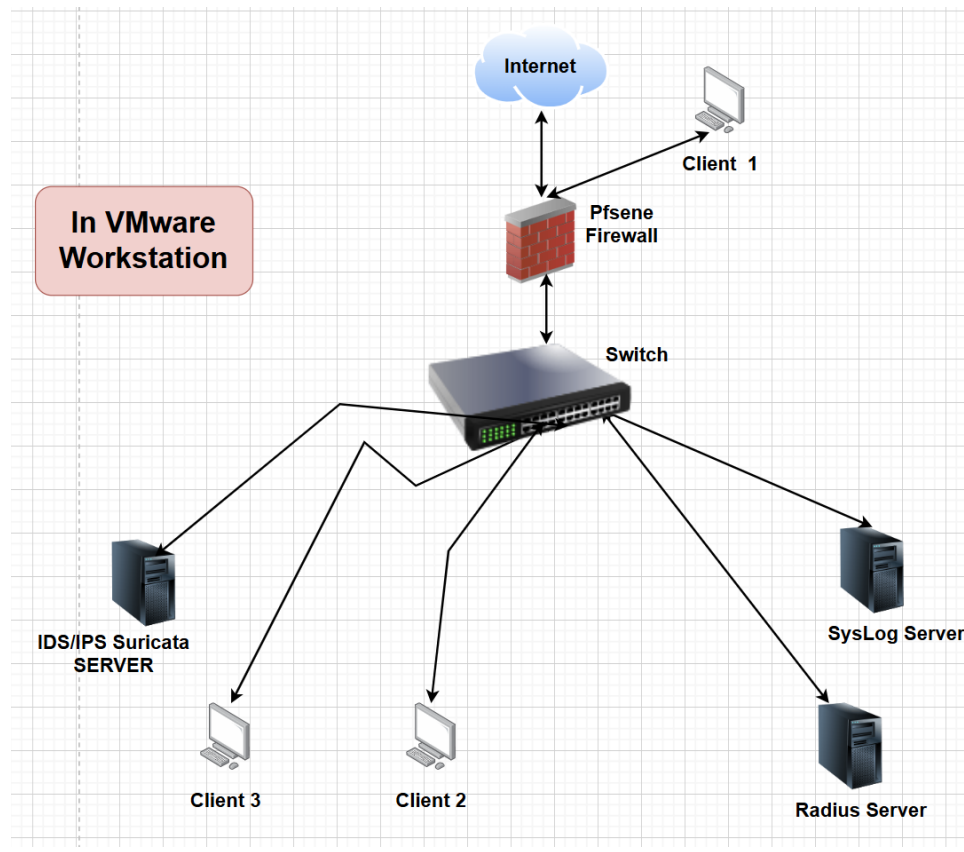


Mini Project in Network Security



Note That:

-  Use VMware to implement the topology mentioned above and verify the connectivity between all devices."
-  The answers must be individual for each student
-  The submission deadline will be during the last week of practical lectures, and it will serve as a replacement for the final exam."
-  Do not copy or look at the IP addresses assigned by your colleagues.

Part 1: Network Design & Components

1. Explain the function of each component in the given topology (Firewall, Switch, Routers, Radius Server, NTP Server, Syslog Server, Clients).
2. What is the purpose of having Radius, NTP, and Syslog servers in a secure network environment?
3. Assign IP addresses (suggested by yourself) to all interfaces, computers, servers, and the firewall.

Part 2: Configuration & Setup

❖ Install and configure pfSense Firewall, then perform the following tasks:

1. Connect the internal LAN network to the external WAN network.
2. Create a user in pfSense Firewall with your name and assign admin privileges to that user.
3. Create a firewall rule to block the ICMP protocol from Client 1 in the WAN network to Client 2 in the internal LAN network, then test it.
4. Create a firewall rule to block the HTTPS protocol from Client 1 in the WAN network to the pfSense Firewall, then test it.

5. Review the traffic logs on the pfSense server.
6. Apply any security settings/configurations you find appropriate to enhance the firewall protection.

❖ **On Routers, apply the following:**

1. Protect all router ports by assigning a password for each port.
2. Change the router hostname to your name.
3. Verify the connectivity between the three routers
4. Apply security settings to limit password cracking attempts (blocking – delay).
5. Configure settings to generate messages/logs for both successful and failed login attempts.
6. Configure SSH access.
7. Create three (3) CLI Views, each containing a set of commands, then create one (1) Super View that includes the previously created two CLI Views.
8. Create different users on Router 3 and assign specific privileges to each user.

❖ **On Router 2, apply the following Port Security configurations:**

Configure Router 3 to function as an NTP server, change its date and time to today's date, and then synchronize all switches and servers with the NTP server. Verify that the synchronization is successful.

❖ **Radius Server**

Install a Radius Server (NPS on Windows Server) and connect it to the switch to secure login using centralized AAA authentication.

❖ **Syslog Server**

Install a Syslog Server and collect event logs from the routers, firewall, servers, and client machines. Extract three security-related events from the logs.

❖ **Install and configure Suricata IDS/IPS Server, then perform the following tasks:**

1. Install Suricata on a dedicated server (Linux VM or Windows).
2. Connect the Suricata server to the network in a way that allows it to monitor traffic passing between the Firewall and the LAN Switch.
3. Configure Suricata with basic rule sets (Emerging Threats or custom rules).
4. Generate traffic from Client 1 (WAN) to Client 2 (LAN) and observe how Suricata detects suspicious activity.
5. Capture and analyze at least two alerts generated by Suricata (e.g., port scanning, suspicious ICMP, or HTTP traffic).
6. Document the Suricata dashboard or log outputs, showing detected events.
7. Apply at least one custom rule (e.g., blocking ICMP echo requests or detecting traffic on a specific port), then test and verify.

❖ **Security Analysis**

1. Perform a network scan using Nmap from Client 2. Identify the open ports on the Radius Server.
2. Use Wireshark on Client 1 to capture traffic between Client 2 and Router 2. Provide:
 - Protocol used
 - Source and destination addresses
 - Whether the data is encrypted or not
3. Relate the results of Nmap and Wireshark with the Firewall rules: How do firewall policies reduce

❖ **Report & Recommendations**

Prepare a **final report** that includes:

1. Screenshots of each configuration step (Firewall, Radius, NTP, Syslog, IDS/IPS Suricata SERVER, Wireshark).
2. A short analysis of potential attacks on this topology.
3. Security recommendations to improve the network setup.

Good luck

Sadeq Al-Soufi